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List of Symbols & Abbreviations

Symbol / Abbreviation	Meaning
AI	Artificial Intelligence
ML	Machine Learning
UI	User Interface
SVM	Support Vector Machine
DNN	Deep Neural Network
LR	Logistic Regression
RF	Random Forest
XGBoost	Extreme Gradient Boosting
BMI	Body Mass Index
TP	True Positive
TN	True Negative
FP	False Positive
FN	False Negative
ROC	Receiver Operating Characteristic
AUC	Area Under Curve
API	Application Programming Interface
ReLU	Rectified Linear Unit
CKD	Chronic Kidney Disease
ECG	Electrocardiogram
IDE	Integrated Development Environment

ABSTRACT

Chronic diseases such as heart disease, diabetes, and chronic kidney disease represent major global health challenges, contributing significantly to mortality and healthcare burden. Early risk assessment is essential for timely medical intervention; however, traditional diagnostic procedures rely heavily on manual clinical evaluation, which can be time-consuming and susceptible to human error. This project, titled “*Multiple Disease Risk Prediction System*,” aims to develop an intelligent machine learning–based solution capable of predicting multiple diseases through automated analysis of patient health parameters.

The proposed system is implemented as a locally hosted web application using Python and the Streamlit framework to provide an interactive and user-friendly interface. The system integrates three independent prediction modules for heart disease, diabetes, and chronic kidney disease within a unified platform. Publicly available medical datasets were utilized, followed by preprocessing steps including missing value handling, feature scaling, and dataset normalization to improve model performance.

A Deep Neural Network architecture with multiple hidden layers and ReLU activation functions was employed for disease prediction. Dropout layers were incorporated to reduce overfitting, while the model was optimized using the Adam optimizer and cross-entropy loss function for binary classification. Model performance was evaluated using standard metrics such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC.

The novelty of the proposed system lies in integrating multiple disease prediction models into a single accessible application designed for non-technical users, enabling rapid health risk assessment without requiring specialized medical expertise. The developed system successfully demonstrates that machine learning techniques can assist in early disease detection and decision support.

In conclusion, the Multiple Disease Risk Prediction System provides a cost-effective, scalable, and efficient approach for preliminary disease risk evaluation. The system can be further enhanced by incorporating additional diseases, real-time clinical data integration, and advanced deep learning techniques to support future intelligent healthcare systems.

1. Introduction

1.1 Background

Over the past few years, chronic illnesses like heart disease, diabetes, and chronic kidney disease have become significant global health issues.

According to global health statistics, these diseases account for a significant proportion of deaths and long-term medical complications worldwide. Rapid urbanization, sedentary lifestyles, unhealthy dietary habits, and increasing stress levels have contributed to the growing prevalence of these conditions.

Traditional disease diagnosis primarily depends on clinical examinations, laboratory tests, and physician expertise. Although effective, these methods can be time-consuming and may delay early detection, particularly in regions where medical resources are scarce. With the progress of Artificial Intelligence and Machine Learning, healthcare systems are increasingly adopting data-driven approaches to support medical decision-making.

Machine learning techniques enable computers to analyse large volumes of medical data, identify hidden patterns, and make predictions based on patient health parameters. These capabilities make machine learning highly suitable for disease risk prediction, assisting healthcare professionals in early diagnosis and preventive care.

1.2 Need for Disease Prediction

Early prediction of diseases is crucial for enhancing patient outcomes and lowering healthcare expenses. Many chronic diseases develop gradually and remain undetected until severe symptoms appear. Timely identification of disease risk allows individuals to adopt preventive measures, lifestyle modifications, and early treatment strategies.

Healthcare systems often face challenges such as increasing patient load, shortage of medical professionals, and delayed diagnostic procedures. Automated disease prediction systems can act as supportive tools by providing quick preliminary assessments based on patient data.

A unified system capable of predicting multiple diseases through a single platform can significantly enhance accessibility and usability. Such systems can assist both medical practitioners and individuals by offering rapid, data-driven health risk evaluation without requiring extensive medical expertise.

1.3 Problem Statement

Despite advancements in healthcare technology, most disease prediction tools focus on a single disease and operate independently. Patients and healthcare providers often need to use separate systems for different medical conditions, leading to inefficiency and fragmented analysis.

Additionally, manual diagnosis processes can be prone to delays and inconsistencies due to human limitations. There is a need for an integrated system that can analyse patient health parameters and predict the likelihood of multiple diseases efficiently and accurately.

Therefore, the problem addressed in this project is the development of an intelligent, machine learning–based system capable of predicting the risk of heart disease, diabetes, and chronic kidney disease through a unified and user-friendly web application.

1.4 Objectives

The primary objectives of the proposed project are:

- To design and develop a Multiple Disease Risk Prediction System using machine learning techniques.
- To analyse medical datasets related to heart disease, diabetes, and chronic kidney disease.
- To perform data preprocessing including missing value handling, feature scaling, and normalization.

- To train and evaluate predictive models using appropriate machine learning and deep learning algorithms.
- To develop an interactive web application using Streamlit for easy user interaction.
- To provide quick and accurate disease risk predictions based on user-input medical parameters.
- To assess model performance using metrics like Accuracy, Precision, Recall, F1-Score, and ROC-AUC.

1.5 Scope of the Project

The scope of this project focuses on building a locally hosted web-based application capable of predicting the likelihood of multiple diseases using machine learning models. The system integrates prediction modules for heart disease, diabetes, and chronic kidney disease within a single interface.

The application is designed primarily for educational and research purposes and provides preliminary risk assessment rather than medical diagnosis. The system operates without requiring cloud infrastructure, external databases, or specialized hardware, making it cost-effective and easy to deploy.

Future enhancements may include integration with real-time hospital databases, mobile application deployment, addition of more disease prediction modules, and incorporation of advanced deep learning techniques for improved prediction accuracy.

1.6 Report Organization

This report is organized into several chapters describing different aspects of the project:

- Chapter 1 presents the introduction, background, objectives, scope, and overview of the project.

- Chapter 2 discusses the literature survey and related research work in disease prediction systems.
- Chapter 3 explains the system design, architecture, datasets, and preprocessing techniques used.
- Chapter 4 describes the methodology, machine learning algorithms, and evaluation techniques employed.
- Chapter 5 details the implementation process, tools, technologies, and system development.
- Chapter 6 presents the experimental results, performance analysis, and discussion.
- Chapter 7 concludes the project and discusses future scope and enhancements.

2. LITERATURE SURVEY

Disease prediction using Machine Learning and Artificial Intelligence has become an active research area in modern healthcare systems. With the rapid growth of medical data and computational power, researchers have developed intelligent systems capable of predicting diseases at an early stage using patient medical records, laboratory results, and lifestyle parameters.

This chapter reviews previous research contributions, existing disease prediction systems, techniques used in earlier studies, and the limitations that motivated the development of the proposed Multiple Disease Risk Prediction System.

2.1 Related Work

Several Scientists have investigated the use of Machine Learning algorithms for predicting diseases.

Early studies focused on statistical analysis methods for detecting diseases such as heart disease and diabetes. Traditional models relied mainly on clinical expertise and manual diagnosis, which were often time-consuming and prone to human error.

Later research introduced Machine Learning models such as:

- Logistic Regression for probability-based disease prediction
- Decision Trees for rule-based classification
- Support Vector Machines (SVM) for high-dimensional medical data analysis
- Random Forest and ensemble learning techniques for improved prediction accuracy

Recent studies demonstrate that machine learning models can successfully analyze large medical datasets to predict diseases before symptoms become severe. Research has

shown that combining multiple algorithms and comparing their performance improves reliability and diagnostic accuracy.

Modern AI-based healthcare research also emphasizes early detection systems capable of assisting doctors in clinical decision-making, reducing diagnostic delays, and improving patient survival rates.

2.2 Existing Medical Prediction Systems

Various medical prediction systems have been developed to assist healthcare professionals.

Some commonly developed systems include:

- Heart Disease Prediction Systems using clinical parameters such as blood pressure, cholesterol level, age, and ECG results.
- Diabetes Prediction Models based on glucose levels, BMI, insulin level, and family history.
- Kidney Disease Prediction Systems utilizing laboratory test results such as serum creatinine, albumin levels, and specific gravity.

Many existing systems are designed to predict only a single disease, requiring separate platforms for different medical conditions. These systems often use web-based or desktop interfaces where users input patient data and receive risk predictions.

Although effective individually, these solutions lack integration and require multiple applications for comprehensive health analysis.

2.3 Techniques Used in Earlier Studies

Earlier disease prediction research employed several computational techniques:

1. Statistical Methods

- Linear Regression

- Bayesian Analysis
- Probability Models

These methods worked well for small datasets but struggled with complex medical relationships.

2. Machine Learning Algorithms

Researchers later adopted machine learning techniques such as:

- Logistic Regression
- Decision Tree Classifiers
- Random Forest Algorithm
- Support Vector Machine (SVM)
- K-Nearest Neighbors (KNN)

These algorithms improved prediction accuracy by learning patterns from historical patient data.

3. Ensemble Learning Approaches

Recent works combine multiple models to achieve better performance and robustness.

4. Data Preprocessing Techniques

- Handling missing values
- Feature scaling
- Feature selection
- Data normalization

Proper preprocessing significantly enhanced model performance in healthcare datasets.

2.4 Limitations of Existing Systems

Despite significant advancements, existing disease prediction systems have several limitations:

1. Single Disease Focus

Most systems predict only one disease, making healthcare monitoring fragmented.

2. Limited Dataset Integration

Many studies rely on small or specific datasets, reducing generalization capability.

3. Lack of User-Friendly Interfaces

Some models remain research prototypes without practical deployment.

4. Data Imbalance Issues

Medical datasets often contain uneven class distribution, affecting prediction accuracy.

5. Interpretability Challenges

Complex machine learning models sometimes lack transparency, making clinical adoption difficult.

6. Absence of Multi-Disease Prediction Platforms

There is a need for integrated systems capable of predicting multiple diseases simultaneously.

2.5 Motivation for Proposed Work

To overcome these limitations, the proposed Multiple Disease Risk Prediction System integrates prediction models for Heart Disease, Diabetes, and kidney disease into a single unified platform. The system aims to provide:

- Early disease risk assessment
- Improved prediction accuracy
- Easy-to-use interface

3. PROPOSED WORK

This chapter describes the overall design and working of the proposed Multiple Disease Risk Prediction System. The system aims to predict the likelihood of multiple chronic diseases using machine learning techniques applied to patient medical data. The application integrates prediction modules for heart disease, diabetes, and chronic kidney disease within a unified web-based interface.

The proposed system combines data preprocessing, machine learning model training, and an interactive user interface to provide fast and reliable disease risk assessment.

3.1 System Architecture

The system follows a modular architecture consisting of frontend, backend, and machine learning components.

The architecture includes the following major components: (**represented in Fig. 3.1**)

1. User Interface (Frontend)

Developed using Streamlit, allowing users to input medical parameters through interactive forms.

2. Data Processing Module

Performs preprocessing operations such as missing value handling, normalization, and feature scaling before prediction.

3. Machine Learning Model Module

Contains trained prediction models for each disease. The models analyze user inputs and generate prediction results.

4. Prediction Engine

Processes input parameters and returns disease risk prediction results in real time.

5. Output Module

Displays prediction results to users in an understandable format.

The system operates locally without external database connections or cloud integration, making deployment simple and cost-effective.

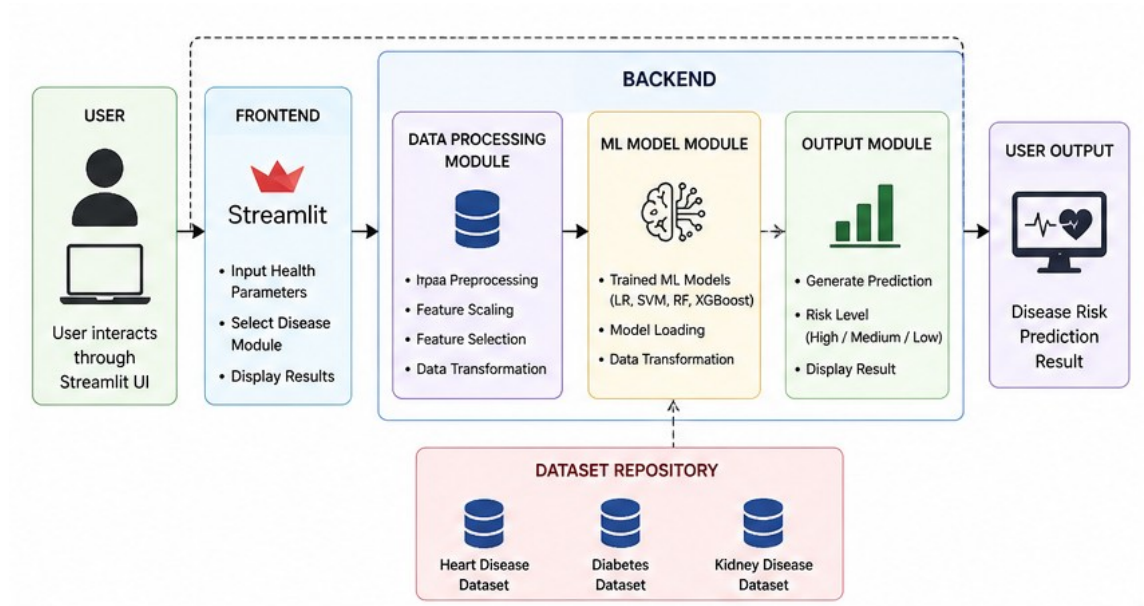


Fig. 3.1- System Architecture of Multiple Disease Risk Prediction System

3.2 System Workflow

The workflow of the proposed system follows sequential steps: **(represented in Fig. 3.2)**

1. User launches the application locally.
2. User selects the disease prediction module.
3. Health parameters are entered through input forms.
4. Data preprocessing is applied automatically.
5. The trained machine learning model processes the input.
6. Prediction result is generated.
7. Disease risk outcome is displayed to the user.

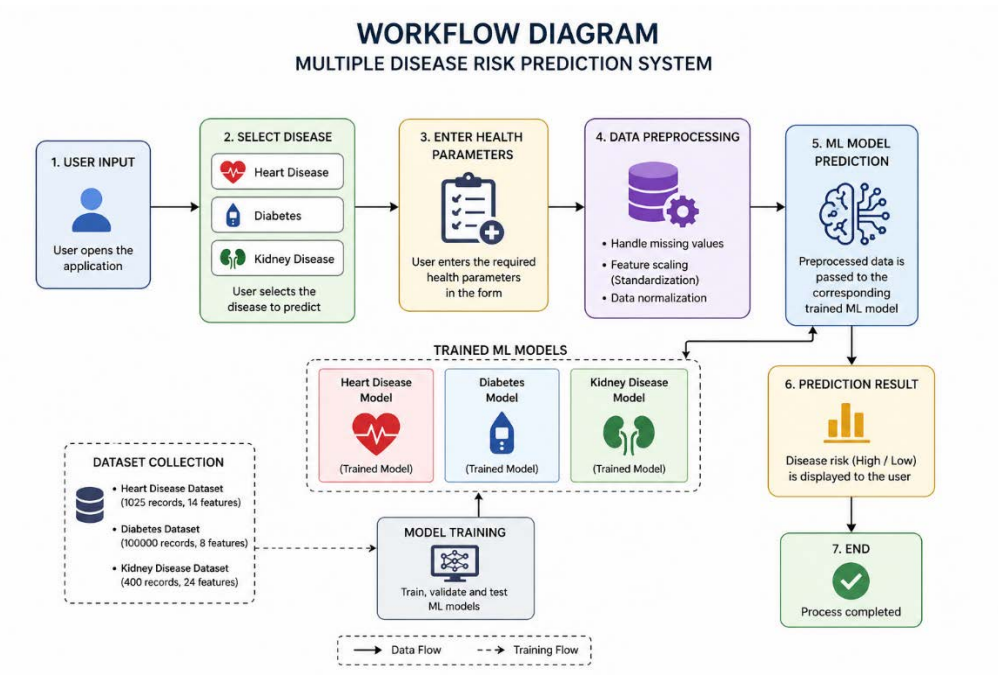


Fig. 3.2- System Workflow Diagram

3.3 Dataset Description

The performance of any machine learning system depends largely on the quality of data used for training. Publicly available healthcare datasets were used in this project for model development.

3.3.1 Heart Disease Dataset

The heart disease dataset is a combination of multiple clinical databases including Cleveland, Hungary, Switzerland, and VA Long Beach datasets.

Dataset Characteristics:

- Number of Records: 1,025
- Number of Features: 14
- Target Variable: Presence or absence of heart disease

Important Features Include:

- Age
- Sex
- Chest Pain Type
- Resting Blood Pressure
- Cholesterol Level
- Fasting Blood Sugar
- Electrocardiographic Results
- Maximum Heart Rate Achieved
- Exercise Induced Angina

This dataset enables prediction of cardiovascular risk based on patient clinical attributes.

3.3.2 Kidney Disease Dataset

The chronic kidney disease dataset contains clinical and laboratory measurements collected from patients over a two-month period at a hospital in Tamil Nadu, India.

Dataset Characteristics:

- Number of Records: 400
- Number of Features: 24
- Target Variable: Chronic Kidney Disease classification

Key Attributes Include:

- Specific Gravity
- Albumin

- Blood Urea
- Serum Creatinine
- Hemoglobin Level
- Blood Pressure
- Red Blood Cell Count

The dataset supports early detection of kidney-related complications.

3.3.3 Diabetes Dataset

The diabetes dataset consists of medical and demographic information collected from patients along with their diabetes diagnosis status.

Dataset Characteristics:

- Number of Records: 100,000
- Number of Features: 8
- Target Variable: Diabetes (Positive/Negative)

Major Features Include:

- Glucose Level
- Body Mass Index (BMI)
- Age
- Blood Pressure
- Insulin Level
- Pregnancies
- Skin Thickness

The dataset helps identify diabetes risk based on physiological parameters.

3.4 Data Preprocessing

Raw medical datasets often contain inconsistencies, missing values, and varying feature scales. Therefore, preprocessing is an essential step before training machine learning models.

The following preprocessing techniques were applied:

1. Handling Missing Values

Missing or null values were replaced using median imputation for numerical attributes to maintain data consistency.

2. Data Cleaning

Irrelevant or duplicate records were removed to improve dataset quality.

3. Feature Scaling

Standard Scaling was applied to normalize feature ranges, ensuring equal contribution of all attributes during model training.

4. Train-Test Split

Datasets were divided into training and testing sets to evaluate model generalization capability.

3.5 Feature Selection

Feature selection plays a critical role in improving prediction accuracy and reducing model complexity.

The following approaches were used:

- Correlation analysis to identify strongly related features
- Removal of redundant and low-impact attributes
- Selection of medically relevant parameters based on domain knowledge

Feature selection helps:

- Reduce overfitting
- Improve training efficiency
- Enhance model interpretability
- Increase prediction performance

Only the most significant clinical features were retained for training the machine learning models.

4. METHODOLOGY

This chapter describes the methodology adopted for developing the Multiple Disease Risk Prediction System. The proposed system utilizes machine learning techniques to analyse patient medical data and predict the likelihood of heart disease, Diabetes and Kidney Disease.

The methodology consists of the data preprocessing features scaling, model training, algorithm selection, and the performance evaluation using standard machine learning metrics.

4.1 Machine Learning Algorithms

Multiple Various machine learning algorithms were applied and evaluated to determine the most effective model for predicting diseases. Each algorithm learns patterns from historical medical data and predicts disease risk based on input parameters.

4.1.1 Logistic Regression

Logistic Regression is supervised Learning algorithm used for binary classification problems. It estimates the probability of occurrence of an event by fitting data to a logistic function.

In disease prediction, Logistic Regression calculates the probability of whether a patient is likely to have a disease or not.

Advantages:

- Simple and efficient.
- Works well for the linearly separable data.
- Easy interpretation of results.

Mathematical Representation:

$$p(y = 1) = \frac{1}{1 + e^{-z}} \quad (4.1)$$

Where: $z = wX + b$

4.1.2 Support Vector Machine (SVM)

The Support Vector Machine is an effective classification tool that distinguishes data points by employing an optimal hyperplane, which maximizes the separation margin between various classes.

SVM is effective for the high- dimensional datasets commonly found in medical applications.

Key Features:

- Works well with complex datasets.
- Uses kernel functions for non-linear classification.
- Robust against overfitting.

4.1.3 Random Forest

Random Forest is a type of ensemble learning method that creates several decision trees during the training phase and makes predictions by using a majority vote approach.

Each tree is trained using random subsets of data and features, which improves prediction accuracy and reduces overfitting.

Advantages:

- High accuracy.

- Handles missing values effectively.

4.1.4 XGBoost (Extreme Gradient Boosting)

XGBoost is a sophisticated ensemble learning algorithm that utilizes gradient boosting methods. It constructs decision trees in a sequence, with each subsequent tree addressing the mistakes made by the ones before it.

XGBoost utilizes second-order gradient optimization and regularization techniques for improved performance.

Advantages:

- High predictive power.
- Faster computation.
- Built-in regularization to prevent overfitting.
- Supports parallel processing.

4.2 Feature Scaling

Medical datasets often contain features with different numerical ranges. For example, age values may range from 20–80 while cholesterol values may exceed 300. Such variation can negatively affect algorithm performance.

To address this issue, Standard Scaling was applied.

Standardization transforms features using:

$$Z = \frac{X - \mu}{\sigma} \quad (4.2)$$

Where:

- X = Original Feature Value.
- μ = Mean of Features.

- σ = Standard Deviation.

Feature scaling ensures:

- Faster model convergence.
- Equal contribution of all features.
- Improved prediction accuracy.

4.3 Training Process

The model training pipeline followed these steps:

1. Dataset collection from publicly available sources.
2. Data cleaning and handling missing values.
3. Feature selection based on the correlation analysis.
4. Feature Scaling using StandardScaler.
5. Splitting dataset into training and testing sets.
6. Training and machine learning models.
7. Model validation using testing data.
8. Saving trained models for deployment in Streamlit application.

The trained models were loaded during runtime to generate real-time predictions.

4.4 Model Evaluation Metrics

To evaluate model performance, several standard classification metrics were used.

4.4.1 Accuracy

Accuracy measures the proportion of correctly predicted instances among total predictions.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (4.3)$$

Where:

- TP = True Positive.
- TN = True Negative.
- FP = False Positive.
- FN = False Negative.

4.4.2 Precision

Precision tells us how many predicted positive cases are actually positive.

$$precision = \frac{TP}{TP + FP} \quad (4.4)$$

High precision means fewer false positive predictions.

4.4.3 Recall

Recall measures the ability of the model to correctly identify the true or actual positive cases.

$$recall = \frac{TP}{TP + FN} \quad (4.5)$$

High recall is important in healthcare to avoid missing disease cases.

4.4.4 F1- Score

F1-Score is the harmonic mean of the precision and recall providing a balanced evaluation metric.

$$F1 \text{ Score} = 2 \times \frac{precision \times recall}{precision + recall} \quad (4.6)$$

It is especially useful when datasets are imbalanced.

4.5 Summary

The methodology combines preprocessing techniques, multiple machine learning algorithms, and performance evaluation strategies to develop an accurate disease prediction system. Comparing multiple algorithms enables selection of the best-performing model for each disease module, ensuring reliable and efficient predictions within the proposed application.

5. IMPLEMENTATION

This chapter describes the practical implementation of the Multiple Disease Risk Prediction System. The system integrates machine learning models with a web-based interface to provide real-time disease risk prediction. The implementation involves model development, application deployment, and user interaction through an intuitive interface.

5.1 Tools and Technologies

The following tools and technologies were used during development:

- Programming Language used: Python.
- Machine Learning Libraries: Scikit-Learn, NumPy, Pandas.
- Data Visualization: Matplotlib
- Web Framework: Streamlit
- Development Environment: Jupyter Notebook / Google Colab
- Model Storage: Pickle files
- Operating System: Windows

Python was selected due to its extensive and vast support for machine Learning and data analysis libraries.

5.2 Software Requirements

The software requirements for implementing the system are:

- Python 3.x
- Streamlit Framework
- Scikit-learn Library

- Pandas and NumPy Libraries
- Matplotlib
- Web Browser (Chrome/Edge)
- IDE such as Jupyter Notebook or VS Code

All tools used are open-source and freely available.

5.3 Hardware Requirements

The system does not require specialized hardware and can operate on a standard computer system.

Minimum Hardware Configuration:

- Processor: Intel core i3 higher.
- RAM: 4 GB minimum (8 GB recommended).
- Storage: 12 GB free disk space.
- Internet Connection (for dataset download only)

5.4 Model Development

Model development involved multiple stages:

1. Dataset collection from publicly available repositories.
2. Data preprocessing includes handling missing values and scaling features.
3. Feature selection based on correlation analysis.
4. Training multiple machine learning models and algorithms:
 - o Logistic Regression.
 - o Support Vector Machine (SVM).

- o Random Forest.
 - o XGBoost.
- 5. Model evaluation using performance metrics.
- 6. Selection of best-performing model.
- 7. Saving trained models using Pickle for deployment.

Currently, the heart disease prediction module is fully developed and integrated into the application.

5.5 User Interface

The user interface was developed using Streamlit, allowing interaction without traditional frontend development.

Interface Features:

- Simple input forms for health parameters
- Disease selection option
- Instant prediction output
- User-friendly layout designed for non-technical users

The interface workflow:

1. User opens the application.
2. Selects disease module.
3. Inputs medical parameters.
4. Clicks Predict button.
5. Prediction result displayed instantly.

5.6 Code Flow

The system follows the following execution flow: **(represented in Fig. 5.1)**

1. Load trained machine learning model.
2. Accept user input from Streamlit form.
3. Convert input into numerical format.
4. Apply preprocessing and scaling.
5. Pass processed data to trained model.
6. Generate prediction result.
7. Display disease risk outcome.

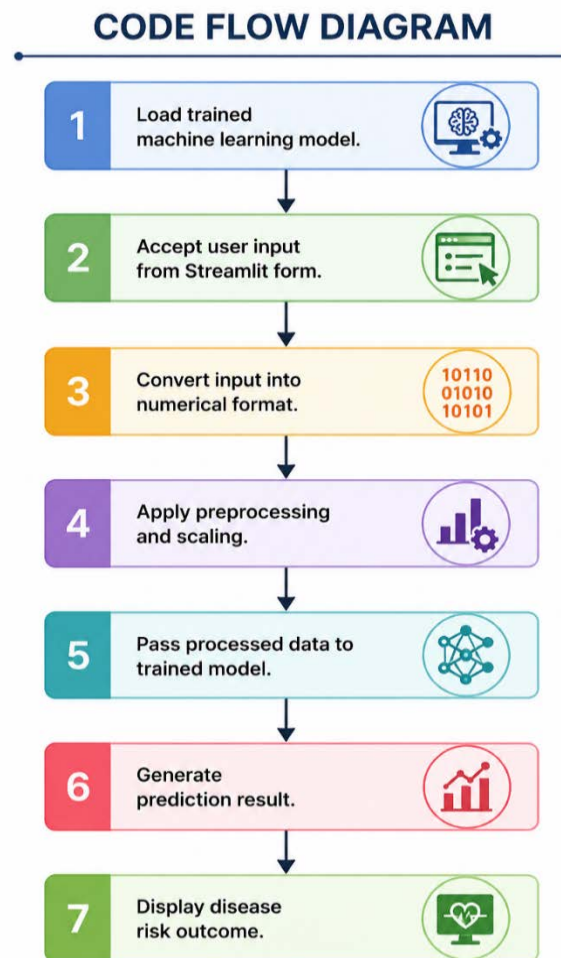


Fig. 5.1 – Code Flow Diagram

6. Results

6.1 Experimental Results

The system was tested using unseen test datasets to evaluate model performance. Predictions were generated using trained Machine Learning models for disease classification.

The results demonstrate that the machine learning algorithms can effectively predict disease risk using patient health parameters.

6.2 Accuracy Comparison

Different algorithms were compared based on prediction accuracy.

Algorithm	Accuracy (%)
Logistic Regression	79%
SVM	68%
Random Forest	98%
XGBoost	94%

The comparison helped Identify the best performing algorithm for each disease module.

6.3 Model Performance

Performance of the models was conducted using following:

- Accuracy.
- Precision.
- Recall.
- F1 – Score.
- ROC-AUC Curve.

The selected models showed strong classification capability and reliable prediction performance.

6.4 Screenshots of System

Includes the following screenshots:

- Application Registration Page (**represented in Fig 6.1.**)
- Disease Selection Page (**represented in Fig. 6.2.**)
- Input Parameter Form (**represented in Fig. 6.3.**)
- Prediction Result Output (**represented in Fig. 6.4.**)

Fig 6.1-Application Registration page

Multi Disease Risk Prediction System

Yash Pandey Patient ID: 19783 System ID 25 Years Age Male Gender 2026-04-27 Visit Date

Select Disease

☒ Heart Disease ☐ Diabetes ☐ Kidney Disease

Fig. 6.2-Disease Selection Page

Heart Disease Prediction

Valid Age Range: 16 - 90 Years

Age	Cholesterol	Exercise Induced Angina
30 - +	200 - +	Yes
Gender	Fasting Blood Sugar > 120 mg/dl	Oldpeak
Male	True	1.00 - +
Chest Pain Type	Rest ECG	ST Slope
Typical Angina	Normal	Upsloping
Resting BP	Max Heart Rate	Number of Major Vessels (0-4)
120 - +	150 - +	0 - +
		Thalassemia
		Normal

Predict

Fig. 6.3-Input Parameter Form

Heart Disease Prediction

Valid Age Range: 16 - 90 Years

Age	Cholesterol	Exercise Induced Angina
30 - +	200 - +	Yes
Gender	Fasting Blood Sugar > 120 mg/dl	Oldpeak
Male	True	1.00 - +
Chest Pain Type	Rest ECG	ST Slope
Typical Angina	Normal	Upsloping
Resting BP	Max Heart Rate	Number of Major Vessels (0-4)
120 - +	150 - +	0 - +
		Thalassemia
		Normal

Predict

☒ Low Risk of Heart Disease

Fig. 6.4-Prediction Result Output

6.5 Discussion

The experimental results confirm that machine learning models can assist in early disease risk prediction. Ensemble models like Random Forest and XGBoost have enhanced performance because they can effectively capture intricate relationships. medical datasets.

The system successfully provides fast predictions through a user-friendly interface, making it suitable for educational demonstrations and preliminary health assessment.

7. CONCLUSIONS

The Multiple Disease Risk Prediction System demonstrates the effectiveness and the working of machine learning techniques in healthcare applications. The developed system provides accurate, fast, and user-friendly disease risk prediction using clinical parameters.

The project highlights how intelligent systems can assist healthcare professionals by enabling early diagnosis and preventive decision-making. The successful implementation confirms the feasibility of integrating machine learning into real-world healthcare support systems.

7.1 Project Achievements

The project successfully achieved the following:

- Developed a machine learning–based disease prediction system.
- Integrated multiple disease prediction modules.
- Implemented preprocessing and feature engineering techniques.
- Designed an interactive Streamlit web application.
- Generated real-time disease risk predictions.

Demonstrated practical application of AI in healthcare.

7.2 Learning Outcomes

During the project, the team gained knowledge in:

- Machine Learning model development.
- Data preprocessing techniques.
- Feature engineering.
- Model evaluation and comparison.

- Web application deployment using Streamlit.
- Healthcare data analytics.

The project enhanced both technical and problem-solving skills.

8. SCOPE FOR FUTURE WORK

The proposed Healthcare system can be further enhanced in several ways:

- The system can be extended to predict additional diseases such as liver disease, lung disease, cancer risk, and hypertension.
- Integration with hospital databases and electronic health records can enable real-time clinical decision support.
- Developing an Android or iOS application would improve accessibility and allow remote health monitoring.
- Advanced deep learning models and larger datasets can improve prediction accuracy and support complex medical analysis.
- Hosting the application on cloud platforms would allow multi-user access and real-world deployment.

References

- [1] D. Dua and C. Graff, “UCI Machine Learning Repository,” University of California, Irvine, School of Information and Computer Sciences, 2019.
- [2] J. Smith, “Heart Disease Dataset,” Kaggle Dataset, 2020. [Online]. Available: <https://www.kaggle.com/datasets/johnsmith88/heart-disease-dataset>
- [3] M. T. Mustafa, “Diabetes Prediction Dataset,” Kaggle Dataset, 2021. [Online]. Available: <https://www.kaggle.com/datasets/iammustafatz/diabetes-prediction-dataset>
- [4] UCI Machine Learning Repository, “Chronic Kidney Disease Dataset,” University of California Irvine, 2015. [Online]. Available: <https://archive.ics.uci.edu/dataset/336/chronic+kidney+disease>
- [5] T. Chen and C. Guestrin, “XGBoost: A Scalable Tree Boosting System,” in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016.
- [6] L. Breiman, “Random Forests,” *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [7] C. Cortes and V. Vapnik, “Support-vector networks,” *Machine Learning*, vol. 20, pp. 273–297, 1995.
- [8] D. W. Hosmer, S. Lemeshow, and R. X. Sturdivant, *Applied Logistic Regression*. Hoboken, NJ, USA: Wiley, 2013.
- [9] F. Pedregosa *et al.*, “Scikit-learn: Machine Learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [10] Streamlit Inc., “Streamlit documentation,” 2023. [Online]. Available: <https://docs.streamlit.io>

APPENDIX

Appendix A — Source Code

This section includes important parts of the implementation code used for developing the Multiple Disease Risk Prediction System.

Example contents:

- Model training script. (**represented in Fig. A.1**)
- Streamlit application code. (**represented in Fig. A.2**)
- Model prediction functions. (**represented in Fig. A.3**)

```
test_size = 0.20
X_train_full, X_test, y_train_full, y_test = train_test_split(
    X, y, test_size=test_size, stratify=y, random_state=42
)

X_train, X_val, y_train, y_val = train_test_split(
    X_train_full, y_train_full, test_size=0.20, stratify=y_train_full, random_state=42
)

imputer = SimpleImputer(strategy="median")
X_train_imp = imputer.fit_transform(X_train)
X_val_imp = imputer.transform(X_val)
X_test_imp = imputer.transform(X_test)

scaler_candidates = {
    "StandardScaler": StandardScaler(),
    "RobustScaler": RobustScaler(),
    "QuantileTransformer": QuantileTransformer(
        n_quantiles=max(10, min(100, len(X_train_imp))),
        output_distribution="normal",
        random_state=42
    )
}
```

Fig. A.1 – Model Training Code.

```

disease_type = st.radio(
    "Select Disease",
    ["Heart Disease", "Diabetes", "Kidney Disease"],
    horizontal=True
)

st.markdown("---")

if disease_type == "Heart Disease":

    st.subheader("❤️ Heart Disease Prediction")
    st.info("Valid Age Range: 16 - 90 Years")

    col1, col2, col3 = st.columns(3)

    with col1:
        age = st.number_input("Age", 16, 90, 30)
        gender = st.selectbox("Gender", ["Male", "Female"])
        chest_pain_help = """
        **Identify your chest pain:**
        * **Typical Angina:** Squeezing or heavy pressure during activity.
        * **Atypical Angina:** Sharp poking or shortness of breath.
        * **Non-anginal:** Muscle-like pain or pain when breathing deeply.
        * **Asymptomatic:** No pain felt at all.
        """
        chest_pain = st.selectbox(
            "Chest Pain Type",
            ["Typical Angina", "Atypical Angina",
            "Non-anginal Pain", "Asymptomatic"], help=chest_pain_help
        )

```

Fig. A.2 – Streamlit Application Code.

```

def get_probabilities(model, X_data):
    if hasattr(model, "predict_proba"):
        return model.predict_proba(X_data)[: , 1]
    score = model.decision_function(X_data)
    return 1.0 / (1.0 + np.exp(-score))

def metric_row(y_true, prob, threshold):
    pred = (prob >= threshold).astype(int)
    try:
        auc = roc_auc_score(y_true, prob)
    except Exception:
        auc = 0.0
    return {
        "threshold": float(threshold),
        "accuracy": accuracy_score(y_true, pred),
        "precision": precision_score(y_true, pred, zero_division=0),
        "recall": recall_score(y_true, pred, zero_division=0),
        "f1": f1_score(y_true, pred, zero_division=0),
        "roc_auc": auc,
        "pred": pred
    }

```

Fig. A.3 - Model prediction functions.

Appendix B — Dataset Sample

This appendix provides sample records from datasets used in the project.

Include:

- Heart Disease dataset sample rows

age	sex	cp	trestbps	chol	fb	restecg	thalach	exang	oldpeak	slope	ca	thal	target
52	1	0	125	212	0	1	168	0	1	2	2	3	0
53	1	0	140	203	1	0	155	1	3.1	0	0	3	0
70	1	0	145	174	0	1	125	1	2.6	0	0	3	0
61	1	0	148	203	0	1	161	0	0	2	1	3	0

- Diabetes dataset sample rows

id	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr	bu	sc	sod	pot
0	48	80	1.02	1	0		normal	notpresen	notpresen	121	36	1.2		
1	7	50	1.02	4	0		normal	notpresen	notpresent		18	0.8		
2	62	80	1.01	2	3	normal	normal	notpresen	notpresen	423	53	1.8		

Appendix C — User Manual

System Usage Steps

1. Install required Python libraries.
2. Run the Streamlit application using command:

```
streamlit run app.py
```
3. Open the local web application in browser.
4. Select disease prediction module.
5. Enter required health parameters.
6. Click Predict.
7. View disease risk result.